Self-Defence Legislation Preferences: Evidence from Discrete Choice Modeling

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ECO499H1: Honours Essay in Applied Microeconomics

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April 23rd, 2026

Abstract

Self-defence legislation varies across countries in how it weighs the well-being of defenders and attackers. "Make My Day" laws in some U.S. states grant immunity to individuals who use any level of force against someone who unlawfully enters their residence, prioritizing the defender's well-being. Canada's Criminal Code S.34 requires defensive force be proportionate to the threat posed, balancing both parties' well-being. We categorize these approaches as unrestricted self-defence legislation (USDL) and proportionate self-defence legislation (PSDL), respectively. Importantly, some people strongly prefer USDL over PSDL and vice-versa, but what demographic characteristics are associated with these preferences? We examined how self-construal, perceived safety, and institutional trust were associated with these preferences. 91 participants completed an online survey: they answered items that measured these demographic characteristics and ranked, by preference, communities that varied in crime rate and self-defense legislation. We used an exploded logit model on these ranked preferences to uncover the associations between these demographic characteristics and preferences. We found that higher independent self-construal predicted greater preference for USDL across both crime types, and that such individuals accepted higher crime rates in exchange for USDL. Contrary to expectations, interdependent self-construal did not predict preference for PSDL. Those feeling less safe preferred USDL, as did those with greater trust in police, courts, and government.

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Introduction

In the early morning of August 18th 2025, a crossbow-wielding man allegedly broke into an apartment in Lindsay, Ontario. As a result, the tenant of the apartment awoke and found the alleged home intruder. A scuffle between the two individuals allegedly ensued where the tenant used a knife to defend himself against the home invader. This altercation resulted in the alleged home invader sustaining life-threatening injuries: he had to be airlifted to a Toronto hospital to receive life-saving treatment. After an investigation by the Kawartha Lakes police, they arrested and charged the tenant with aggravated assault and assault with a weapon (Daigle, 2025). Once this event was shared online by news stations, many politicians were unhappy with this judicial outcome. Federal Conservative Leader Pierre Poilievre stated that "The law needs to be clear ... If someone invades your home, you have the right to defend your home and your family" (Daigle, 2025). Moreover, Ontario Premier Doug Ford commented "I know if someone breaks into my house or someone else's, you're going to fight for your life ... This guy has a weapon ... you're going to use any force you can to protect your family" (Daigle, 2025). Some media outlets mentioned that this event would not have happened in some U.S. states due to legislation inspired by castle doctrine ("Make My Day" law), which provide immunity from prosecution for individuals who use deadly force against someone who unlawfully and forcibly enters a person's residence (Ibrahimpoor, 2025; Lev, 2025; National Conference of State Legislatures, 2025). Moreover, this legislation can grant immunity from the criminal charges being laid altogether, meaning the matter may never reach trial. Pierre Poilievre proposed that Canada should have a similar legislative approach when handling self-defence cases (Steven, 2025). In Canada, if you defend yourself in a manner that constitutes a criminal offense, the police can still charge you with a criminal offense. It is up to the criminal courts to determine whether you are guilty of said criminal offense. To be found not guilty, criminal courts require a claimant of self-defence to have responded to

the threat in a reasonable and proportionate manner. More specifically, under Criminal Code S.34, you are not guilty of an offense if (a) you believe on reasonable grounds that force is being used against you or another person or that a threat of force is being made against you or another person, (b) the act that constitutes the offence is committed for the purpose of defending or protecting yourself or the other person from that use or threat of force, and (c) the act committed is reasonable in the circumstances (Branch, 2025). Theoretically speaking, Criminal Code S.34 sounds very reasonable and ideal as it ensures that the defender and the attacker are not disproportionately harmed. However, past research has found that individuals facing an imminent threat do not have the time or cognitive capacity to adjust their level of force in a reasonable and proportionate manner, resulting in disproportionate force being applied (Nieuwenhuys, 2012; Renden, 2017). Moreover, past research has shown that even well-trained law enforcement officers make lethal force errors when subjected to high stress simulations (Baldwin, 2022). Canadian courts attempt to account for these dynamics when determining if a claimant of self-defence should be acquitted of a criminal offense (Gilmore, 2023).

In the context of intruding a lawful occupant's residence, the "Make My Day" legislation and Criminal Code S.34 weigh the well-being of the lawful occupant and the intruder differently. In particular, "Make My Day" legislation prevents criminal charges from being laid on the lawful occupant and allows said occupant to use any level of force on the intruder, which suggests that the lawful occupant's well-being is prioritized over the intruder's well-being. On the other hand, Criminal Code S.34 does not prevent criminal charges from being laid on the lawful occupant and requires the occupant to use an amount of force that is proportionate to the threat imposed by the intruder, which implies that the lawful occupant's and intruder's well-being are more balanced relative to the "Make My Day" legislation. In essence, each type of legislation maximizes the well-being of the intruder and the lawful occupant with a different set of weights. We were interested in determining what demographic characteristics led to such weighting preferences.

Literature Review

In 1994, a prominent cultural psychologist named Theodore Singelis created a scale to determine how people view themselves in relation to others, known as the self-construal scale. He argued that people differ in how they view themselves in relation to others. For example, some people may view themselves as distinct from other people in their social group(s) whereas other people may view themselves as similar to other people in their social group(s). Broadly speaking, self-construal can be thought of as an intricate combination of thoughts, feelings, and actions concerning one's relationship with other people. Importantly, self-construal has two dimensions: independent self-construal and interdependent self-construal.

Independent self-construal is defined as how much a person sees themselves as separate, unique, and independent from others. Individuals with high independent self-construal prefer to communicate directly, express their uniqueness, and promote their personal goals. In addition, they focus on their personal abilities, thoughts, and feelings. Therefore, when tasked with thinking about themselves, they think about their own characteristics and do not refer to their social group(s)'s characteristics. Moreover, they do not consider how their own characteristics overlap with their social group(s)'s characteristics. For example, suppose that an individual with high independent self-construal spends most of their time with a rowing club, and, in their free time, they enjoy reading comic books. When tasked with thinking about themselves, they may highlight how their goals and characteristics are unique from the rest of their group: they would describe themselves as an athletic person who enjoys reading comic books. When tasked with thinking about others, they acknowledge other people's unique characteristics independent of how they relate to their own characteristics. At a high level, independent self-construal captures the need for uniqueness and separateness commonly found in Western cultures. As a result, this subscale covaries with individualism (Akaliyski et al., 2026). Finally, this type of self-construal is measured by averaging an individual's

responses across 15 statements that they rate on a Likert scale from one to seven (one = strongly disagree, 7 = strongly agree). For example, individuals with high independent self-construal would typically strongly agree with the following statements: "I enjoy being unique and different from others in many respects" and "I do my own thing, regardless of what others think," and strongly disagree with the following statements: "I usually go along with what others want to do, even when I would rather do something different" and "If my brother or sister fails, I feel responsible."

Interdependent self-construal refers to how much a person sees themselves as connected, similar, and interdependent with others. People with high interdependent self-construal care about how their social group(s) perceive them through statuses, roles, and relationships. In addition, they focus on belonging and fitting in social group(s), properly occupying their own place, and being able to indirectly communicate with other people. Therefore, when tasked with thinking about themselves, these individuals think of the characteristics they share with their social group(s). For example, suppose that a person with high interdependent self-construal spends the majority of their time with a rowing club and enjoys knitting. When tasked with thinking about themselves, they will consider how their characteristics converge with the rest of their rowing club. Perhaps, they would describe themselves as an athletic person who loves to row. When tasked with thinking about others, they consider how people's characteristics within social group(s) are interconnected with each other. Also, individuals with high interdependent self-construal gain utility from maintaining harmonious relationships with other people; tools, such as indirect communication, help reach this objective. At a high level, interdependent self-construal emphasizes the desire for connectedness and relations found in non-Western cultures. Therefore, it covaries with collectivism (Akaliyski et al., 2026). Finally, this type of self-construal is measured by averaging an individual's responses across 15 statements that they rate on a Likert scale from one to seven. For example, individuals with high interdependent self-construal would typically strongly agree with "Even when I strongly

disagree with group members, I avoid an argument" and "I feel good when I cooperate with others,", and strongly disagree with "I act the same way no matter who I am with" and "Being able to take care of myself is a primary concern for me."

Crucially, despite independent and interdependent self-construal being conceptually separate, they can coexist within an individual. Triandis (1989) provides the theoretical foundation for such a claim. He theorized that within each person, there exist three types of selves. The private self encapsulates cognitions (conscious thoughts or mental representations) about how a person describes their traits, states, or behaviours (e.g., "I am happy," "I am extroverted"). The public self concerns cognitions about how a person thinks they are perceived by the general public (e.g., "People think that I am sad," "People think that I am introverted"). The collective self encompasses the cognitions concerning how a person thinks they are perceived by some social groups (e.g., "My family thinks that I am happy," "My rowing club thinks that I am strong"). When thinking about the self, Triandis (1989) theorized that individuals would sample their cognition(s) from one of the three types of selves. Crucially, these sampling probabilities are determined by immediate situational factors and how much these selves have developed over time. For example, people from individualistic cultures would be pushed to sample more from the private self whereas people from collectivistic cultures would sample more from the public and collectivistic self. Therefore, individualistic cultures would develop the private self while collectivistic cultures would develop the public and collectivistic selves. However, some individuals may be exposed to both types of cultures, fostering well-developed private, public, and collectivistic selves. Moreover, immediate circumstantial factors may push an individual to sample more from one self over others independent of culture exposure. For example, suppose that an individual is asked to think about how their goals diverge from the rest of their social group, then they may be more likely to sample from the private self even if they were raised in a collectivistic culture. Trafimow et al. (1991) provide empirical evidence for such a claim. They recruited people from collectivistic and individualistic

countries: China and North America, respectively. They randomly assigned these individuals to one of two primers: in the first condition, participants were instructed to think about how they are different from friends and family; in the second condition, participants were asked to think about how they were similar to friends and family. They had these individuals complete 20 sentences that began with the words "I am." Then, they assessed whether these sentences were made in relation to the self (i.e., private self cognition: personal qualities, attitudes, or beliefs), or others (i.e., collectivistic self cognition: social groups or demographic categories). They found that the type of culture and prime significantly affected the type of cognition produced: Chinese people, on average, had more collectivistic cognitions than North American people, and North American people, on average, had more individualistic cognitions than Chinese people. Independent of culture, individuals who were primed with an individualistic stimulus had more private self cognitions. Additionally, individuals who were primed with a collectivistic stimulus had more collective self cognitions. In essence, these findings suggest that each individual has private and collectivistic selves and that the retrieval of a cognition from one of these selves is dependent on culture and circumstantial factors. Singelis (1994) posited a relationship between self-construal and these three selves; he argued that people with high independent self-construal would be more likely to sample from the private self and that people with high interdependent self-construal would be more likely to sample from the public and collective selves. Knowing that being equally exposed to individualistic and collectivistic cultures fosters well-developed private, public, and collectivistic selves, independent and interdependent self-construal can coexist within an individual. Moreover, Cross (1995) provides evidence for such a claim. She measured independent and interdependent self-construal in East Asian exchange students and American students during their first year of American graduate school. She found no significant difference in independent self-construal between the East Asian students and American students ($M_{EastAsians}$ = 4.85, SD = .63; $M_{Americans}$ = 4.89, SD=.62), t(148) < 1.0). However, she

found that East Asian students had significantly higher interdependent self-construal than American students ($M_{EastAsians}$ = 4.03, SD = .77; $M_{Americans}$ = 3.4, SD=.78), t(144) = 4.88, p < .001). Crucially, interdependent and independent self-construal were not correlated with each other in either group (r= .02 for East Asians and r=-.07 for Americans), suggesting that these measures represent two unrelated parts of the self. In sum, independent and interdependent self-construal can coexist within an individual despite being conceptually separate from one another.

To construct his independent and interdependent self-construal scales, Singelis (1994) borrowed items from previous self-construal and individualism-collectivism scales (Bhawuk and Brislin, 1992; Cross, 1995; Yamaguchi, 1994). Some items were modified to better fit the self-construal they reflected. Other items were created by Singelis (1994) to directly capture some aspects of independent and interdependent self-construal that were not captured in the aforementioned items (e.g. "Speaking up during a class is not a problem for me" to measure directness and verbal expression). Altogether, Singelis' original scale comprised of 45 items in total. To validate his scale, he recruited 364 students from the University of Hawaii at Manoa (M_{age} = 21.63). The sample was predominantly Japanese (34.1%), followed by Caucasian (13.77%), Chinese (12.0%), other (12.0%), Filipino (8.9%), Hawaiian or part-Hawaiian (7.3%), Korean (3.6%), mixed (5.6%), African American (2.2%), and Samoan (.6%). He presented each participant with a questionnaire where they had to indicate how much they agreed with each self-construal item on a scale from 1 to 7 (1 = strongly disagree, 7 = strongly agree). All items were randomly ordered. After collecting his data, Singelis (1994) performed a principal component analysis. He discarded the items that did not load highly on independent or interdependent self-construal (less than .35) or loaded equally on both types. Twelve items for each type were selected for the final scale. The two factors accounted for 25% of the total variance. A maximum likelihood confirmatory factor analysis showed that the two-factor model provided better fit than a one-factor model. In addition, the Cronbach

alphas for the independent and interdependent self-construal scales were .69 and .73, respectively. These values were adequate given the broadness of the construct and the range of behaviour, thoughts, and feelings assessed by the scale. Finally, the scores for each type of self-construal were uncorrelated with each other ($r = -.044$), supporting the notion that independent and interdependent self-construal are conceptually distinct and that they can coexist within an individual. Similar results were replicated in another sample of students from University of Hawaii at Manoa ($n = 160$). Following the original publication, Singelis revised his scale by adding six items to improve internal reliabilities of the original scale. Thereafter, Cronbach Alpha reliabilities with the 15 items have been ranging from the high .60s to the middle .70s (Singelis, 2007). Finally, other research has defended the validity of his scale (Gudykunst and Lee, 2003; Kim and Raja, 2003).

Previous research has found that individuals with high independent self-construal act more impulsively, competitively, and selfishly (Biddlestone et al., 2020; Magee & Galinsky, 2008; Pan et al., 2024; Xiang et al., 2019). These findings suggest that these individuals primarily factor in their own well-being when making decisions that may affect others. In addition, prior research has reported that individuals with high interdependent self-construal act more deliberately, collaboratively, and ethically (Cojuharenco, et al., 2012; Konrath, Bushman, & Grove, 2009; Utz, 2004; Zhang & Shrum, 2008; Pan et al., 2024). These results suggest that these individuals consider others' well-being when making decisions that may influence others. Moreover, Pan et al. (2024) provide empirical support for such suggestions. They surveyed American ($n = 888$) and Chinese ($n = 844$) citizens during the early stage of the COVID-19 pandemic about their engagement in behaviors that protect their health through prevention. They found that American participants had higher independent self-construal than Chinese participants. Moreover, American participants reported significantly greater engagement in self-focused behaviors than Chinese participants, suggesting that people with high independent self-construal care more about their personal well-being. On the other hand, Chinese participants had

significantly higher interdependent self-construal than American participants. In addition, Chinese participants reported greater engagement in other-involved behaviors than American participants, suggesting that people with high interdependent self-construal care more about collective well-being. Critically, self-construal partially explained the observed differences in engagement in these protective behaviors. In sum, people with high independent self-construal weigh personal well-being more than collective well-being whereas people with high interdependent self-construal weigh collective well-being more than personal well-being. These behavioural insights are important as they may inform preferences with respect to self-defence legislation. We focus on two types of self-defence legislation: proportionate self-defence legislation (PSDL), where the defensive force used must be proportional to the threat posed by the attacker, and unrestricted self-defence legislation (USDL), where any level of force may be used against an attacker. Given that people with high independent self-construal care more about their personal well-being, they may prefer USDL more than PSDL as it allows them to maximize their personal well-being without legal repercussion. On the other hand, people with high interdependent self-construal may prefer PSDL more than USDL as it ensures that collective well-being is maximized.

Other circumstantial factors may also contribute to self-defence legislation preferences. We hypothesize that citizens who feel unsafe in their daily lives may prefer USDL as knowing that they can defend themselves without facing any legal repercussion provides them with a sense of comfort. In addition, we posit that people who do not trust the police, court system, and/or government may prefer the USDL as it gives them the opportunity to uphold their own sense of justice when these governing entities cannot be relied upon (Angell, 2025). Finally, the type of crime may influence self-defence preferences. For example, murders threaten a person's physical well-being more than robberies; therefore, most people would state that self-defence legislation is more important in the context of murders than robberies. In sum, we are interested in determining the

relationship between self-construal, safety, trust, and crime type and self-defence legislation preferences.

In terms of hypotheses, independent of demographic characteristics and self-defence legislation, we predicted that individuals would assign lower utility to higher crime rates (1). Independent of crime type, we hypothesized that people with higher independent self-construal would derive higher utility from USDL than from PSDL (2). In addition, we posited that people with higher interdependent self-construal would extract greater utility from PSDL than USDL (3). Moreover, we postulated that individuals who feel less safe would derive more utility from USDL than from PSDL (4). We hypothesized that individuals who distrust the police, court system, and government more would derive greater utility from USDL than from PSDL (5). We predicted that these preferences would be stronger for communities plagued by murders than communities affected by robberies. (6). In addition, we hypothesized that individuals with higher than average independent self-construal would be willing to have unrestricted self-defence legislation at the expense of a higher overall crime rate (7). For example, they may prefer a community with USDL and 3 murders per year over a community with PSDL and 2 murders per year.

Method

We used an online survey along with a logit demand model to uncover how safety, trust, and independent and interdependent self-construal modulated self-defence legislation and crime rate preferences.

General Survey Design

We presented our 91 participants with an online survey where they indicated their levels of independent and interdependent self-construal, how safe they felt on a daily basis, and how much they trusted the police, courts, and government. They were also asked to imagine themselves moving into a new community of 1000 people for the next twelve years. They were told that this community had a fixed number of crimes each year and that all residents had an equal chance of being a victim. Each participant was presented with six

communities representing all combinations of self-defence legislation type (unrestricted vs. proportionate) and annual crime rate (2, 3, or 4 crimes per year). For each type of crime (i.e., murder or robbery), they had to rank these communities from most preferred to least preferred.

Participants

We recruited 99 participants from Prolific, an online survey-taking platform. 91 of these participants passed all our attention and comprehension checks ($n_{male} = 58$, $n_{female} = 31$). Most of our participants were White (40.7%), West Asian (18.7%), and East Asian (11.0%). With respect to political identity, our participants were mainly centrist (49.5%). Concerning religion, our participants were mostly Muslim (26.4%), agnostic (18.7%), and Christian (16.5%). Regarding country of residence, most of our participants were from Egypt (23.1%), the United Kingdom (18.7%), and the United States (13.2%). See Appendix A for the general demographic information of our participants. All participants were fluent in English. Regarding independent self-construal, our participants had a mean independent self-construal of 4.930 ($s = 0.732$), which roughly corresponded to "somewhat agreeing" with the items of this construct. Similarly, our participants had a mean interdependent self-construal of 4.706 ($s = 0.650$). Our participants had a mean trust of 2.377 ($s = 0.715$), indicating that, on average, they did not trust the police, court system, and government very much. Regarding safety, the mean perceived safety in daily life was 5.745 ($s = 0.798$), suggesting that, on average, participants felt safe. For more detailed information on these demographic characteristics, see the table below. Each participant was paid $6.26 CAD for completing our 15-minute survey. The Economics Undergraduate Research Grant funded by Robert C. Douglas provided the necessary funds for participant compensation.

Descriptive Statistics of Demographic Variables of Interest ($N = 91$)

Variable	N	Mean	SD	Median	Min	Max
Independent Self-Construal	91	4.930	.732	5.000	2.867	6.667
Interdependent Self-Construal	91	4.706	.650	4.733	2.600	6.200
Trust	91	2.377	.715	2.333	1.000	4.000
Safety	91	5.745	.798	5.750	3.500	7.000

Note. INDS and INTS are computed as the mean of 15 items each, rated on a 1–7 Likert scale. Trust and safety are composite averages of 3 and 4 items respectively, rated on 1-4 and 1-7 Likert scales, respectively. Higher values indicate greater independent self-construal, interdependent self-construal, trust in police, courts, and government, and perceived safety.

Materials and Design

To begin, participants completed a consent form where they were informed that (a) they would have to make judgments about a series of hypothetical scenarios that refer to crime, and (b) they would have to provide information about demographic background, beliefs, and attitudes (See Appendix B). Afterward, participants were presented with the demographic information and community ranking tasks in random order to prevent order effects (Lucas, 1992).

Regarding the demographic information task, participants completed the self-construal, safety, trust, and general demographic information tasks in random order. For the self-construal task, the independent and interdependent self-construal items were mixed together and randomly ordered; participants rated how much they agreed or disagreed with each item on a 7-point Likert scale. We used the items from Singelis' (2007) updated scale where each type of self-construal had 15 items. For the perceived safety task, participants mentioned how much they agreed or disagreed with four perceived safety items on a 7-point Likert scale. These items were derived from the Personal Safety Perception Scale and were randomly ordered (Dabbous et al., 2025). For the trust task, participants stated how much confidence they had in the police, court system, and government separately on a 4-point Likert scale; these items were drawn from the World Values Survey

(Haerpfer et al., 2020). For the general demographic information section, we asked participants for their gender, age, ethnicity, political identification, religious beliefs and country of residence. See Appendix C for more details about these questions. Within these demographics questions, we had two attention check questions: they asked participants to enter a specific number.

Concerning the community ranking task, participants completed the general introduction first: They were asked to imagine themselves moving into a new home in a small village where they would be living for the next 12 years. They were informed that this village had a population of 1000 people, but it was unusual as the number and type of crime were exactly the same each year. They were also told that each crime had a single attacker and victim and that each resident had an equal chance of being a victim of these crimes. Finally, they were presented with a picture of the village (See Appendix D for more information). The hypothetical 12-year tenure ensured that participants' stated preferences reflected what they would prioritize in practice when selecting a long-term residence, boosting external validity. We chose a population of 1000 people as it (a) made victimization plausible, and (b) enabled participants to trade between crime rate and self-defence legislation type. Earlier, we posited that individuals with high independent self-construal would be willing to have unrestricted self-defence legislation at the expense of a higher overall crime rate (7). Importantly, if the population size were too small (e.g., 20 people), then the probability of being victimized would be very high, rendering the crime rate very salient. Given that most people have a strong desire for self-preservation, most participants would exclusively rank the communities by crime rate to increase their chances of survival (Karni and Schmeidler, 1986; Brown et al., 1999). Such a design would dramatically reduce the likelihood of finding evidence that supports hypothesis (7). Moreover, at least 81% of the world's population lives in communities of size greater than 5000 (United Nations, 2025); therefore, such trade-off between crime rate and self-defence legislation may exist in real-world preferences. Furthermore, we indicated that the crimes

were intentional. If we did not specify this information, perhaps some participants would have thought that some crimes were accidental (e.g., deadly car crash). In that case, the connection between crime rate and self-defence legislation becomes murkier. One explanation lies in the human threat management system. Throughout human evolutionary history, humans have been exposed to intentional threats of physical harm (Chagnon, 1988). Therefore, humans have developed a precautionary self-protection system that detects features in others that connote the possibility of intentional harm and responds to the perception of these features with the activation of the fight or flight response (Neuberg et al., 2011). A similar system was not developed for unintentional threats of physical harm, implying that people are more concerned about defending themselves from intentional threats than unintentional ones (Neuberg et al., 2011). Therefore, self-defence and its legislation are typically interpreted in the context of intentional harm. We mentioned that each crime had one attacker and one victim so that participants could conceivably engage in some acts of self-defence, which would increase the relevance of self-defence legislation. We specified that every resident had an equal chance of being a victim of these crimes to ensure that all participants (a) were presented with the same probability of victimization and (b) did not find any ways to remove themselves from the threat imposed by crime through unaccounted factors. For example, suppose that we did not specify that everybody had an equal chance of being a victim of these crimes and that one of our participants was very muscular. Conceivably, they would believe that these crimes would not affect them personally as their muscular build would deter attackers; therefore, they might focus on how the self-defence legislation affects the overall population and not themselves. Ultimately, this threat removal may lead them to select the proportionate self-defence legislation. This intuition is supported by Crandall's (1978) research, which found that a personal threat typically produces increased concern for the self along with decreased concern for the interests of others, which reduces manifestations of altruistic behavior (i.e. caring for the overall population's well-being by selecting

proportionate self-defence legislation). We included a picture of the village as we wanted our participants to visualize the village's physical attributes in the same manner. In addition, the picture informed the participants that the residents were living close to each other so there was little opportunity for physical evasion; therefore, participants would have to resort to self-defence, rendering self-defence legislation more salient.

Each participant completed the murder ranking and robbery ranking tasks in random order. First, they were presented with a general introduction. They were told that it is important to understand self-defence laws. Then, the unrestricted and proportionate self-defence laws were explained. The former was described as being allowed to use any level of force to defend yourself against an attacker. It was accompanied by an example: "you are allowed to kill an attacker with a firearm if they punched you." The latter was described as being allowed to defend yourself, but the level of force you use must be proportionate to the threat posed by the attacker. It was followed by an example: "if an attacker punches you, it would be illegal to stab them using a knife." Then, they completed a self-defence legislation comprehension question. For example, we asked them whether it would be legal for them to stab an attacker with a knife if the attacker punched them under proportionate self-defence law. Afterward, participants were asked to rank communities that varied across all combinations of self-defence legislation type (unrestricted vs. proportionate) and annual murder or robbery rate (2, 3, or 4 murders or robberies per year). These communities were randomly ordered. We chose rankings as our input as the interpretation is the same across all participants. Moreover, it forces participants to directly compare each community's attributes to other communities, yielding rich data. We chose crime rates of two, three, and four to increase the odds of gathering evidence that supports hypothesis (7). Our literature review informed us that people with high INDS may prefer restrictive self-defence legislation. In addition, previous research states that most people, including the high INDS group, strongly prefer low crime rates (Brenig and Proeger, 2018; Cohen et al., 2004). If the lower bound on crime rate was

high (i.e. 10 crimes per year instead of two), then the high INDS group would be more likely to exclusively rank these communities by crime rate. Setting the lower bound at a small value circumvents this issue: having a lower crime rate renders the crime rate less salient and allows the high INDS group to trade higher crime for unrestricted self-defence legislation. In a similar vein, if the difference in crime rate across communities were very large, then the high INDS group would exclusively rank these communities by crime rate as well. However, if the difference in crime rate across these communities were small (i.e., one-unit), then these people would be more willing to trade higher crime for unrestricted self-defence legislation. See Appendix D for more details about our survey design.

Model

We used an exploded logit model with maximum likelihood estimation to recover how safety, trust, and independent and interdependent self-construal modulated self-defence legislation and crime rate preferences.

To begin, the utility that participant n acquires from community i is specified as:

$$U_{ni} = \boldsymbol{\beta}(\mathbf{D}_n)^\top \mathbf{X}_i + \epsilon_{ni},$$

where $\boldsymbol{X}_i$ represents the attributes of community i[1], $\boldsymbol{D}_n$ reflects the demographic characteristics of participant n, and $\epsilon_{ni} \sim$ i.i.d. standard Type I extreme value.

$$\mathbf{X}_i = \begin{bmatrix} \text{self-defence legislation} \\ \text{number of annual crimes} \end{bmatrix}, \qquad \mathbf{D}_n = \begin{bmatrix} \text{independent self-construal} \\ \text{interdependent self-construal} \\ \text{trust in the police, courts, and government} \\ \text{safety} \end{bmatrix}.$$

Crucially, we allow preference heterogeneity across participants by letting the

[1] USDL was coded as 1 and PSDL was coded as 0

coefficient vector $\boldsymbol{\beta}(\boldsymbol{D}_n)$ depend on participant n's demographic characteristics:

$$\boldsymbol{\beta}(\mathbf{D}_n) = \boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{D}_n,$$

where

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}, \qquad \boldsymbol{\Gamma} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & \gamma_{14} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & \gamma_{24} \end{bmatrix}.$$

In this case, $\boldsymbol{\alpha}$ represents participants' shared baseline preference for self-defence legislation and number of annual crimes. $\boldsymbol{\Gamma}$ reflects the notion that participants' demographic characteristics can influence the utility acquired from self-defence legislation and annual crime rate. It is composed of all possible interaction effects between community attributes $\boldsymbol{X}_i$ and demographic characteristics $\boldsymbol{D}_n$. For example, γ_{11} is the interaction effect between self-defence legislation and independent self-construal.

To make sure that our model is dimensionally consistent, note that $\boldsymbol{\alpha} \in \mathbb{R}^2$, $\boldsymbol{\Gamma} \in \mathbb{R}^{2\times 4}$, and $\mathbf{D}_n \in \mathbb{R}^4$, so $\boldsymbol{\beta}(\mathbf{D}_n) \in \mathbb{R}^2$ as required. Substituting into the utility specification yields

$$U_{ni} = \boldsymbol{\alpha}^\top \mathbf{X}_i + (\boldsymbol{\Gamma}\mathbf{D}_n)^\top \mathbf{X}_i + \epsilon_{ni} = \boldsymbol{\alpha}^\top \mathbf{X}_i + \mathbf{D}_n^\top \boldsymbol{\Gamma}^\top \mathbf{X}_i + \epsilon_{ni}.$$

Linear-in-parameters representation.

We seek to rewrite this model in a more compact form for more succinct notation

later on. Define

$$\mathbf{Z}_{ni} = \begin{bmatrix} D_{n1}X_{i1} \\ D_{n1}X_{i2} \\ D_{n2}X_{i1} \\ D_{n2}X_{i2} \\ D_{n3}X_{i1} \\ D_{n3}X_{i2} \\ D_{n4}X_{i1} \\ D_{n4}X_{i2} \\ X_{i1} \\ X_{i2} \end{bmatrix} \in \mathbb{R}^{10}, \qquad \boldsymbol{\delta} = \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5 \\ \delta_6 \\ \delta_7 \\ \delta_8 \\ \delta_9 \\ \delta_{10} \end{bmatrix} \in \mathbb{R}^{10}.$$

Then the utility function can be written compactly as

$$U_{ni} = \boldsymbol{\delta}^\top \mathbf{Z}_{ni} + \epsilon_{ni}.$$

The correspondence between $\boldsymbol{\delta}$ and the original parameters is

$$\delta_{2k-1} = \gamma_{1k}, \qquad \delta_{2k} = \gamma_{2k}, \qquad k = 1, \ldots, 4,$$

and

$$\delta_9 = \alpha_1, \qquad \delta_{10} = \alpha_2.$$

To see why this notation works, observe that

$$
\begin{aligned}
U_{ni} &= \boldsymbol{\alpha}^\top \mathbf{X}_i + \mathbf{D}_n^\top \boldsymbol{\Gamma}^\top \mathbf{X}_i + \epsilon_{ni} \\
&= \alpha_1 X_{i1} + \alpha_2 X_{i2} + \sum_{k=1}^{4} D_{nk}\Big(\gamma_{1k} X_{i1} + \gamma_{2k} X_{i2}\Big) + \epsilon_{ni} \\
&= \sum_{k=1}^{4}\Big(\gamma_{1k}\, D_{nk} X_{i1} + \gamma_{2k}\, D_{nk} X_{i2}\Big) + \alpha_1 X_{i1} + \alpha_2 X_{i2} + \epsilon_{ni} \\
&= \boldsymbol{\delta}^\top \mathbf{Z}_{ni} + \epsilon_{ni}.
\end{aligned}
$$

Ranking model

Thus far, we have a functional form for the utility U_{ni} that participant n receives from community i. Unfortunately, we did not directly observe this utility value so we cannot directly regress U on $\boldsymbol{Z}$. For each participant n, we observed their preference ranking. Fortunately, our exploded logit model can provide us with the probability of observing participant n's preference ranking under a certain coefficient vector $\boldsymbol{\delta}$.

Suppose participant n is presented with a set of six communities

$$C_n = \{x_1, x_2, x_3, x_4, x_5, x_6\},$$

and provides a complete ranking

$$x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6.$$

The ranking is interpreted as a sequence of choices.

At stage m, the participant chooses item x^m from the remaining choice set

$$C_n^{x^m} = C_n \setminus \{x^1, \ldots, x^{m-1}\}.$$

Under the random utility model with i.i.d. extreme value errors, the probability of choosing alternative i from a choice set C is

$$P(i \mid C) = \frac{\exp(\boldsymbol{\delta}^\top \mathbf{Z}_{ni})}{\sum_{j \in C} \exp(\boldsymbol{\delta}^\top \mathbf{Z}_{nj})}.$$

Taking logs yields:

$$\ln P(i \mid C) = \boldsymbol{\delta}^\top \mathbf{Z}_{ni} - \ln \sum_{j \in C} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{ni}}$$

Therefore, the probability of observing ranking $x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6$ is:

$$L_n\Big(x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6 \mid \boldsymbol{\delta}\Big) = \prod_{m=1}^{5} \frac{\exp(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^m})}{\sum_{j \in C_n^{x^m}} \exp(\boldsymbol{\delta}^\top \mathbf{Z}_{nj})}$$

See Kilani, K. (2025) for the proof of this likelihood function.

Likelihood function

So far, we know the function that dictates the probability of observing participant n's ranking. Now, we want to maximize the probability of observing all participants' rankings by modulating $\boldsymbol{\delta}$.

Let ζ denote the sample size. The likelihood across all participants is

$$\mathcal{L}(\boldsymbol{\delta}; \text{rankings}) = \prod_{n=1}^{\zeta} L_n\Big(x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6 \mid \boldsymbol{\delta}\Big).$$

The parameter vector $\boldsymbol{\delta}$ is estimated using maximum likelihood:

$$\hat{\boldsymbol{\delta}} = \arg\max_{\boldsymbol{\delta}} \mathcal{L}(\boldsymbol{\delta}; \text{rankings}).$$

Taking logs yields the sample log-likelihood

$$\hat{\ell}(\boldsymbol{\delta}) = \sum_{n=1}^{\zeta} \ln L_n\Big(x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6 \mid \boldsymbol{\delta}\Big)$$

$$= \sum_{n=1}^{\zeta} \sum_{m=1}^{5} \left(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^m} - \ln \sum_{j \in C_n^{x^m}} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{nj}} \right)$$

$$= \sum_{n=1}^{\zeta} \Bigg[\left(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^1} - \ln \sum_{j \in C_n^{x^1}} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{nj}} \right) + \left(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^2} - \ln \sum_{j \in C_n^{x^2}} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{nj}} \right) + \left(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^3} - \ln \sum_{j \in C_n^{x^3}} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{nj}} \right)$$

$$+ \left(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^4} - \ln \textstyle\sum_{j \in C_n^{x^4}} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{nj}} \right) + \left(\boldsymbol{\delta}^\top \mathbf{Z}_{nx^5} - \ln \textstyle\sum_{j \in C_n^{x^5}} e^{\boldsymbol{\delta}^\top \mathbf{Z}_{nj}} \right) \Bigg]$$

which is maximized numerically. In essence, this estimation method allows participants to have preference heterogeneity with respect to the community attributes $\mathbf{X}_i$; however, the mapping from demographic characteristics $\mathbf{D}_n$ to preferences $\boldsymbol{\beta}(\boldsymbol{D}_n)$ is the same across all participants. Also, we assume that all participants share a common baseline for $\boldsymbol{\beta}(\boldsymbol{D}_n)$, being $\boldsymbol{\alpha}$.

Results

Survey Results

For the murder ranking task, 61.54% of participants ranked the community with USDL and two yearly murders in first place out of the six communities. 32.97% of participants ranked the community with PSDL and two yearly murders in first place out of the six communities. For the robbery ranking task, 47.25% of participants ranked the community with PSDL and 2 yearly robberies in first place out of the six communities. 45.05% of participants ranked the community with USDL and 2 yearly robberies in first place out of the six communities. For more information, see the tables below.

Probability of Ranking Community First — Murder Task ($N = 91$)

Community	Count	Percentage
Unrestricted law, 2 murders	56	61.54%
Unrestricted law, 3 murders	1	1.10%
Proportionate law, 4 murders	0	0.00%
Proportionate law, 2 murders	30	32.97%
Proportionate law, 3 murders	1	1.10%
Unrestricted law, 4 murders	3	3.30%
Total	91	

Note. Each cell reports the number and share of participants who ranked that community as most preferred.

Probability of Ranking Community First — Robbery Task ($N = 91$)

Community	Count	Percentage
Unrestricted law, 2 robberies	41	45.05%
Unrestricted law, 3 robberies	0	0.00%
Proportionate law, 4 robberies	2	2.20%
Proportionate law, 2 robberies	43	47.25%
Proportionate law, 3 robberies	3	3.30%
Unrestricted law, 4 robberies	2	2.20%
Total	91	

Note. Each cell reports the number and share of participants who ranked that community as most preferred.

Estimation Results

Concerning regression results, recall that our regression assumes that demographic characteristics $\mathbf{D}_n$ shift preferences in the same manner across all participants and that all participants share a common preference baseline for crime rate and self-defence legislation. In addition, the demographic variables were standardized before running our regressions. Regarding the murder ranking task, holding all else equal, participants gained 0.8129 utility

transitioning from PSDL to USDL. All else held constant, participants lost 1.3876 utility from a one-unit increase in the yearly murder rate. All else held constant, participants with perceived safety one standard deviation below the mean gained 0.3586 utility from switching PSDL to USDL relative to participants with mean perceived safety. All else held constant, participants with trust one standard deviation below the mean lost 0.2585 utility from switching PSDL to USDL relative to participants with mean trust. All else held constant, participants with trust one standard deviation above the mean lost 0.2046 utility from a one-unit increase in the yearly murder rate relative to participants with mean trust. All else held constant, participants with INDS one standard deviation above the mean gained 0.5468 utility moving from PSDL to USDL relative to participants with mean INDS.

Murder Task ($N = 91$)

Parameter	**Coef.**	**SE**	***t***	***p***	**Sig.**
γ_{11} (INDS × law)	0.5468	0.1383	3.953	0.0001	***
γ_{21} (INDS × crimes)	0.0600	0.1036	0.579	0.5626	
γ_{12} (INTS × law)	0.0473	0.1351	0.350	0.7263	
γ_{22} (INTS × crimes)	0.0134	0.0972	0.138	0.8905	
γ_{13} (trust × law)	0.2585	0.1380	1.873	0.0611	.
γ_{23} (trust × crimes)	-0.2046	0.0995	-2.055	0.0398	*
γ_{16} (safety × law)	-0.3596	0.1337	-2.689	0.0072	**
γ_{26} (safety × crimes)	0.0492	0.0970	0.507	0.6120	
α_1 (baseline law)	0.8129	0.1251	6.496	0.0000	***
α_2 (baseline crimes)	-1.3876	0.0937	-14.803	0.0000	***
Optimization did converge, McFadden R^2: 0.274					

Note. *** $p < .001$, ** $p < .01$, * $p < .05$, . $p < .10$. All $\mathbf{D}_n$ variables are standardised to mean zero and unit variance prior to estimation. Law is coded as 1 = unrestricted, 0 = proportionate.

Based on these estimated coefficients, participants were most likely to rank the

community with USDL and 2 yearly murders in first place across all one standard deviation demographic shifts. Interestingly, shifting INDS one standard deviation above the mean was associated with being most likely to rank the community with USDL and 3 yearly murders in second place.

Predicted Most Likely Community Rankings — Murder Task

Profile	**Most Likely First Rank**	P	**Most Likely Second Rank**	P
Baseline	U, 2 murders	0.5280	P, 2 murders	0.4962
INDS +1 SD	U, 2 murders	0.5959	U, 3 murders	0.3909
INTS +1 SD	U, 2 murders	0.5335	P, 2 murders	0.4839
Trust +1 SD	U, 2 murders	0.5983	P, 2 murders	0.5103
Safety +1 SD	U, 2 murders	0.4594	P, 2 murders	0.5400

Note. Probabilities are computed as $P(i \mid C) = \exp(\hat{\boldsymbol{\delta}}^\top \mathbf{Z}_{ni}) / \sum_{j \in C} \exp(\hat{\boldsymbol{\delta}}^\top \mathbf{Z}_{nj})$ evaluated at the stated participant profile. Each demographic shift holds all other $\mathbf{D}_n$ variables at their standardised mean of zero. Regarding the computation of the probabilities for the second rank, for each row, we removed the community that was most likely to be ranked first and recomputed the choice probabilities.

Regarding the robbery ranking task, all else held constant, participants lost 1.0240 utility from a one-unit increase in the yearly robbery rate. All else held constant, participants with perceived safety one standard deviation below the mean gained 0.4199 utility from switching PSDL to USDL relative to participants with mean perceived safety. All else held constant, participants with trust one standard deviation above the mean gained 0.3124 utility transitioning from PSDL to USDL relative to participants with mean trust. All else held constant, participants with INDS one standard deviation above the mean gained 0.7250 utility moving from PSDL to USDL relative to participants with mean INDS.

Robbery Task ($N = 91$)

Parameter	Coef.	SE	t	p	Sig.
γ_{11} (INDS × law)	0.7250	0.1383	5.242	0.0000	***
γ_{21} (INDS × crimes)	-0.0079	0.0862	-0.092	0.9271	
γ_{12} (INTS × law)	0.1357	0.1366	0.994	0.3203	
γ_{22} (INTS × crimes)	0.1218	0.0877	1.389	0.1650	
γ_{13} (trust × law)	0.3124	0.1374	2.273	0.0230	*
γ_{23} (trust × crimes)	-0.0392	0.0849	-0.462	0.6439	
γ_{16} (safety × law)	-0.4199	0.1356	-3.097	0.0020	**
γ_{26} (safety × crimes)	-0.1212	0.0829	-1.461	0.1440	
α_1 (baseline law)	0.0357	0.1228	0.291	0.7713	
α_2 (baseline crimes)	-1.0240	0.0805	-12.724	0.0000	***
Optimization did converge, McFadden R^2: 0.1888					

Note. *** $p < .001$, ** $p < .01$, * $p < .05$, . $p < .10$. All $\mathbf{D}_n$ variables are standardised to mean zero and unit variance prior to estimation. Law is coded as 1 = unrestricted, 0 = proportionate.

Based on these estimated coefficients, across most demographic shifts, participants were more likely to rank the community with USDL and 2 yearly robberies in first place. Interestingly, shifting safety by one-standard deviation above the mean was associated with being more likely to rank the community with PSDL and 2 yearly robberies over the community with USDL and 2 yearly robberies in first place.

Predicted Most Likely Community Rankings — Robbery Task

Profile	Most Likely First Rank	P	Most Likely Second Rank	P
Baseline	U, 2 robberies	0.3420	P, 2 robberies	0.5015
INDS +1 SD	U, 2 robberies	0.4595	P, 2 robberies	0.3972
INTS +1 SD	U, 2 robberies	0.3456	P, 2 robberies	0.4450
Trust +1 SD	U, 2 robberies	0.4002	P, 2 robberies	0.4711
Safety +1 SD	P, 2 robberies	0.4191	U, 2 robberies	0.4913

Note. Probabilities are computed as $P(i \mid C) = \exp(\hat{\boldsymbol{\delta}}^\top \mathbf{Z}_{ni}) / \sum_{j \in C} \exp(\hat{\boldsymbol{\delta}}^\top \mathbf{Z}_{nj})$ evaluated at the

stated participant profile. Each demographic shift holds all other $\mathbf{D}_n$ variables at their standardised mean of zero. Regarding the computation of the probabilities for the second rank, for each row, we removed the community that was most likely to be ranked first and recomputed the choice probabilities.

Discussion

As a preface, our regression coefficients assume that demographic characteristics $\mathbf{D}_n$ shift preferences in the same manner across all participants and that all participants share a baseline preference for crime rate and self-defence legislation preferences. In support of hypothesis (1), we found that individuals assigned lower utility to higher crime rates across both crime types ($\alpha_{2,murders} = -1.3876$, $\alpha_{2,robberies} = -1.0240$). Notably, participants lost more utility from a one-unit increase in the yearly murder rate relative to a one-unit increase in the yearly robbery rate. These findings are consistent with murder being deemed more serious than the loss of material goods as the former threatens their physical well-being more than the latter (Thurstone, 1927). Consistent with hypothesis (2), our results indicate that individuals with higher levels of INDS preferred USDL over PSDL across both crime types ($\gamma_{11,murders} = 0.5468$, $\gamma_{11,robberies} = 0.7250$). Not accounting for baseline preference ($\alpha_{1,murders} = 0.8129$), individuals with higher independent self-construal gained more utility transitioning from PSDL to USDL in the context of robberies than murders ($\gamma_{11,robberies} = 0.7250$, $\gamma_{11,murders} = 0.5468$). These results suggest that people with higher INDS generally prefer to protect their personal well-being at the possible expense of the attacker's well-being. In general, individuals assigned more utility to USDL than PSDL in the case of murders as murders strongly threatened their physical well-being ($\alpha_{1,murders} = 0.8129$). No such baseline preference was observed in the robbery task as it posed a lesser threat to their physical well-being. Interestingly, people with higher INDS preferred USDL over PSDL in the case of robberies ($\gamma_{11,robberies} = 0.7250$). One possible explanation is that people with higher INDS focus on autonomy and the prioritization of personal goals, interests, and abilities (Magee & Galinsky, 2008; Park and Levine, 1999;

Singelis, 1994). Therefore, they will be more motivated to protect themselves, even at the attacker's expense, when facing threats to these personal attributes. Contrary to hypothesis (3), individuals with higher interdependent self-construal did not assign greater utility to PSDL than USDL. Again, previous research has found that individuals with higher interdependent self-construal act more deliberately, collaboratively, and ethically, especially with people who are part of their social group(s) (Cojuharenco, et al., 2012; Konrath, Bushman, & Grove, 2009; Utz, 2004; Zhang & Shrum, 2008). However, if they are threatened by an individual, then they are less likely to perceive this individual as being part of their social group(s) (Miller et al., 2010). Therefore, their proclivity to behave more deliberately, collaboratively, and ethically may dissolve, resulting in them not wanting to consider the aggressor's well-being. In accordance with hypothesis (4), participants who felt less safe in their daily lives assigned greater utility to USDL relative to PSDL ($\gamma_{16,murders} = -0.3596$, $\gamma_{16,robberies} = -0.4199$). Incongruent with hypothesis (5), individuals who distrusted the police, court system, and government more derived less utility from USDL than from PSDL ($\gamma_{13,murders} = 0.2585$, $\gamma_{13,robberies} = 0.3124$). Perhaps individuals with lower institutional trust prefer PSDL as this legislation prevents these governing entities from misusing USDL on their citizens. Consistent with hypothesis (6), when accounting for baseline preferences α_1 and α_2, the magnitude of the preferences in hypotheses (1), (2), (4), and (5) were stronger in the murder ranking task than the robbery ranking task. Finally, in line with hypothesis (6), we found that individuals with high independent self-construal were willing to have USDL at the expense of a higher overall crime rate. In the context of murders, our model predicts that individuals with a one-standard deviation increase in INDS would rank a community with 3 yearly murders and USDL over a community with 2 yearly murders and PSDL for second place. However, this pattern was not observed in the robbery task, possibly due to murder imposing greater threat on physical well-being than robberies. In practice, individuals with higher independent self-construal may have an even stronger willingness to accept higher crime

rates in exchange for USDL. We assume that individuals with higher INDS are sensitive to the crime-to-population ratio when trading between crime rate and self-defence legislation. Specifically, a lower ratio, indicating less crime relative to population, increases their willingness to accept higher crime rates in exchange for USDL. In our experiment, participants were presented with crime-to-population ratios of 0.002, 0.003, and 0.004. In 2023, Canada's crime-to-population ratios were drastically lower: 0.0000199 and 0.0005898 for murders and robberies, respectively (Statista, n.d.; Statistics Canada, 2025). Therefore, given these lower ratios, people with higher independent self-construal may be more willing to incur higher crime rates for USDL. Unrelated to our hypotheses, in the context of robberies, our model predicted that individuals with a one-standard deviation increase in perceived safety would rank a community with 2 yearly robberies and PSDL over a community with 2 yearly robberies and USDL for first place. In terms of explanation, individuals who feel safer in their daily lives tend to fear crime less (Dabbous et al., 2025) and to view police as more effective and legitimate (Syropoulos, 2024). For robbery specifically, a crime that poses less immediate physical threat than murder, such individuals may be more comfortable accepting proportionate sentencing, trusting that law enforcement can adequately manage these issues without necessitating harsher deterrents, such as USDL.

Regarding policy implications, Canadians, in general, should assign greater utility to USDL than PSDL across both crime types. Currently, 62% of Canadians believe that crime has been rising in their communities over the last five years (Angus Reid Institute, 2026), whilst only 5% believe there's been a decrease. From these findings, we infer that, on average, Canadians feel more unsafe in their daily lives. In addition, on average, Canadians exhibit trust in police, court system, and government that is higher than our sample mean: 65%, 49%, and 50% of Canadians reported having a good or great deal of confidence in the police, court system, and federal government, respectively (Statistics Canada, 2024). However, Canadians have an average level of independent self-construal

similar to our sample mean (Cai et al., 2025; Kuo et al., 2004; Sato and Cameron, 1999). Altogether, on average, Canadians may gain utility in transitioning from PSDL to USDL.

Should Canada engage in such a legislative transition? Would this change in self-defence legislation decrease crime rates? To answer the latter, Yakubovich et al. (2021) reviewed 25 studies that examined the effects of laws expanding civilian rights to use deadly force in self-defence on crime rates. They found that these laws were associated with no change to small increases in violent crime (total and firearm homicide, aggravated assault, robbery) on average across states. This review, while informative, is subject to a major limitation that complicates the validity of its findings. This literature review claims to examine the impacts of laws expanding civilian rights to use deadly force in self-defence on crime rates. However, when defining whether a state passed such legislation, most of its reference papers rely on Cheng and Hoekstra's (2013) list of states that have passed stand-your-ground (SYG) legislation (Cheng and Hoekstra, 2013; Crifasi et al., 2018; Gius, 2016; Guettabi and Munasib, 2018; Munasib et al., 2018; Levy et al., 2020). This list is problematic as Cheng and Hoekstra (2013) define SYG legislation as legislation that removes the duty to retreat in some place outside the home, which is a legal principle that requires individuals to attempt to retreat or remove themselves from a threatening situation before resorting to force, or deadly force, in self-defence (National Conference of State Legislatures, 2025). It does not mention an expansion to use deadly force in self-defence. This mismatch in definitions obfuscates the impact of laws expanding civilian rights to use deadly force legislation on crime rates. In practice, the legislative change that most closely resembles a transition from PSDL to USDL is the implementation of "Make My Day" laws, which provides immunity from prosecution for individuals who use deadly force against someone who unlawfully and forcibly enters their residence. Therefore, to examine the impact of transitioning from PSDL to USDL on crime rate, future research should focus on the impact of "Make My Day" laws on home-related crime rates (National Conference of State Legislatures, 2025). They should use quasi-experimental designs (e.g.,

difference-in-differences, synthetic difference-in-difference) that exploit variation in the timing of policy adoption across states. Altogether, it is uncertain whether Canada should make the transition from PSDL to USDL without knowing its potential impacts on overall crime rates.

Conclusion

We found that higher independent self-construal predicted greater preference for USDL across both crime types, and that such individuals accepted higher crime rates in exchange for USDL. Contrary to expectations, interdependent self-construal did not predict preference for PSDL. Those feeling less safe preferred USDL, as did those with greater trust in police, courts, and government.

In terms of limitations, participation is voluntary and motivated by financial compensation; therefore, our sample may not be representative of the broader population. Therefore, our results should be interpreted as applying to the population of Prolific users who are willing to complete a 15-minute survey about crime for $15. Regarding our model, it is attractive due to its overall simplicity; however, simple models often have a few shortcomings. In particular, the error term ϵ_{ni} for each participant n and item i is an identically and independently distributed standard type 1 extreme value. This independence property means that the unobserved portion of utility for one item i is unrelated to the unobserved portion of utility for another item j. In other words, the error terms are uncorrelated with each other. Stated differently, this idea implies that $\boldsymbol{\beta}(\boldsymbol{D}_n)$ is sufficiently specified, such that the unobserved portion of utility is "white noise." To have shared error terms, we would need there to be unobserved characteristics that are shared across items. This seems implausible given that the communities provided to the participants were solely defined in terms of crime rate and self-defence legislation. Therefore, item-level variation in taste should be mainly attributed to the aforementioned dimensions. However, there could be some unobserved demographic characteristics within each participant that may lead them to value some dimensions more than others across all

items, leading to correlated error terms. For a given participant n, these unobserved demographic characteristics should correlate across items. This problem can be remediated by including a $\boldsymbol{\nu}_n$ error vector in $\boldsymbol{\beta}(\boldsymbol{D}_n)$ (See Appendix E). In addition, participants' preferences $\beta(\boldsymbol{D}_n)$ may shift differently depending on D_n. This problem can be remediated by including a $\boldsymbol{\Upsilon}_n$ error matrix in $\boldsymbol{\beta}(\boldsymbol{D}_n)$. Another problem with the standard logit specification is that it assumes independence of irrelevant alternatives, which is the idea that $\frac{\mathrm{Pr}_{ni}}{\mathrm{Pr}_{nj}}$ does not depend on any alternatives other than i and k. However, we are not interested in substitution patterns. For such a model specification, see Appendix E.

Acknowledgments

Thank you to Robert C. Douglas for making this research possible with his Undergraduate Economics Research Grant. Thank you to Professor Chen for his excellent support and advice. Thank you to Professor Bobonis for his guidance and providing me with the opportunity to complete this highly interdisciplinary research. Thank you to the Department of Economics at the University of Toronto for providing me with a rigorous framework to think about research (ECO475, ECO439, and ECO372 in particular). Thank you to the Department of Psychology at the University of Toronto for refining my writing skills. Thank you to Professor Bloom for letting me borrow his Qualtrics account. Thank you to my ECO499 cohort for their suggestions. Thank you to all of my friends and family who supported me throughout this project.

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Appendix A

Table 1

Demographic Characteristics of Participants (N = 91)

Characteristic	*n*	%
Gender		
Female	31	34.1
Male	58	63.7
Prefer not to say	2	2.2
Race/Ethnicity		
White/European	37	40.7
West Asian/Middle Eastern	17	18.7
East/Southeast Asian	10	11.0
Hispanic/Latina	7	7.7
Other	6	6.6
Black/African Am.	5	5.5
Multiracial	5	5.5
South Asian	4	4.4
Political Identity		
Centrist	45	49.5
Liberal	32	35.2
Conservative	14	15.4
Religion		
Muslim	24	26.4
Agnostic	17	18.7
Protestant	15	16.5
Catholic	8	8.8
Other	5	5.5
Buddhist	2	2.2
Atheist	20	22.0
Country of Residence		
Egypt	21	23.1
United Kingdom	17	18.7
United States	12	13.2
Poland	9	9.9
Canada	8	8.8
Colombia	3	3.3
South Africa	3	3.3
Other ($\leq$2 each)	18	19.8

Note. Multiracial combines three mixed-race responses. "Other" countries include Spain, Germany, Morocco, Chile, France, Italy, Brazil, Ireland, Finland, Portugal, Mexico, and Romania.

Appendix B

Consent Form

* **University of Toronto**
ECO499: Honours Essay Research Course

Title of Project: The Development of Social Behaviour
Principal Investigator: Dr. Yanyou Chen

Study Background
We are conducting this study to examine how individuals reason and make sense of the social world. Your participation in this study is completely voluntary. You are free to decline to participate, to end participation at any time for any reason, or to refuse to answer any individual question. We anticipate that your involvement will require less than 20 minutes, unless otherwise specified.

Procedures
Participation in this study will involve making judgments about a series of hypothetical scenarios that refer to crime. You will also be asked to answer questions about your demographic background, beliefs, and attitudes. We anticipate that your involvement will require less than 15 minutes unless otherwise specified.

Risks and Benefits
Some questions in this study involve hypothetical scenarios that refer to crime (e.g., robbery or murder). These scenarios are not graphic in nature. You may skip any question or withdraw from the study at any time without penalty. Although this study will not provide any direct benefits to our participants, your participation

makes a significant contribution to our understanding of how people reason and make sense of the social world. You will receive the standard rate set by Prolific.

Confidentiality
All your data will be held in confidence. Only the researchers involved in this study and those responsible for research oversight will have access to the information and data acquired from this study. All your responses will be de-identified and linked to a study-specific ID number. Any files that may link your personal information with the responses you provide will be stored separately from the data in a locked cabinet, a password protected device, or the password-protected lab server. All our data will be analyzed and reported as an aggregate. In cases where we make our data files public, all information included will be completely deidentified. None of the information we provide in our reports, publications and presentations can be used to track your or your personal information.

Questions: If you have any questions about this study, you may contact our lab director, Dr. Yanyou Chen (yanyou.chen@utoronto.ca). You may request for a copy of this consent form from our researcher or obtain further information about your rights as a research participant by contacting the University of Toronto's Ethics Review Office at 416-358-6115.

Agreement to Participate:
I have read the above information and understand the material contained in it. I have had the opportunity to have any questions about this study answered. I certify that I am at least 18 years of age and I agree to participate in this study

- ○ I have read the above information and understand the material contained in it. I certify that I am at least 18 years of age and I agree to participate in this study.
- ○ I do not agree to participate in this study, and I wish to exit the survey

Appendix C

All independent and interdependent self-construal items were presented in random order within one block. The safety and trust questions were presented in a random order within their own blocks. Before answering these questions, participants were told that they should answer the following questions as truthfully as possible.

Independent Self-Construal *(Singelis, 1994; Singelis, 2007)*

Scale: 1 = Strongly disagree, 2 = Disagree, 3 = Disagree somewhat, 4 = Neither agree nor disagree, 5 = Agree somewhat, 6 = Agree, 7 = Strongly agree

1. I enjoy being unique and different from others in many respects.
2. I can talk openly with a person who I meet for the first time, even when this person is much older than I am.
3. I do my own thing, regardless of what others think.
4. I feel it is important for me to act as an independent person.
5. I'd rather say "No" directly, than risk being misunderstood.
6. Having a lively imagination is important to me.
7. I prefer to be direct and forthright when dealing with people I've just met.
8. I am comfortable with being singled out for praise or rewards.
9. Speaking up during a class (or a meeting) is not a problem for me.
10. I act the same way no matter who I am with.
11. I value being in good health above everything.
12. I try to do what is best for me, regardless of how that might affect others.

13. Being able to take care of myself is a primary concern for me.

14. My personal identity, independent of others, is very important to me.

15. I act the same way at home that I do at school (or work).

Note. Independent self-construal score (INDS) is computed as the mean of all 15 items above. Higher values indicate greater independent self-construal.

Interdependent Self-Construal *(Singelis, 1994; Singelis, 2007)*

Please answer the following questions as truthfully as possible.

Scale: 1 = Strongly disagree, 2 = Disagree, 3 = Disagree somewhat, 4 = Neither agree nor disagree, 5 = Agree somewhat, 6 = Agree, 7 = Strongly agree

1. Even when I strongly disagree with group members, I avoid an argument.

2. I have respect for the authority figures with whom I interact.

3. I respect people who are modest about themselves.

4. I will sacrifice my self-interest for the benefit of the group I am in.

5. I should take into consideration my parents' advice when making education/career plans.

6. I feel my fate is intertwined with the fate of those around me.

7. I feel good when I cooperate with others.

8. If my brother or sister fails, I feel responsible.

9. I often have the feeling that my relationships with others are more important than my own accomplishments.

10. I would offer my seat in a bus to my professor (or my boss).
11. My happiness depends on the happiness of those around me.
12. I will stay in a group if they need me, even when I am not happy with the group.
13. It is important to me to respect decisions made by the group.
14. It is important for me to maintain harmony within my group.
15. I usually go along with what others want to do, even when I would rather do something different.

Note. Interdependent self-construal score (INTS) is computed as the mean of all 15 items above. Higher values indicate greater interdependent self-construal.

Perceived Safety *(Dabbous et al., 2025)*

Please indicate how much you agree or disagree with each of the following statements.

Scale: 1 = Strongly disagree, 2 = Disagree, 3 = Disagree somewhat, 4 = Neither agree nor disagree, 5 = Agree somewhat, 6 = Agree, 7 = Strongly agree

1. I feel safe when walking alone during the night.
2. I feel safe when walking alone during the day.
3. I feel safe in places that I usually frequent.
4. I feel safe at home.

Note. Safety score is computed as the mean of all four items. Higher values indicate greater perceived safety.

Institutional Trust *(Haerpfer et al., 2020)*

I am going to name a number of organizations. For each one, could you tell me how much confidence you have in them: is it a great deal of confidence, quite a lot of confidence, not very much confidence or none at all?

Scale: 1 = None at all, 2 = Not very much, 3 = Quite a lot, 4 = A great deal

1. The police
2. The courts
3. The government

Note. Trust score is computed as the mean of all three items. Higher values indicate greater institutional trust.

General Demographics

Participants were kindly asked to fill out this general demographics form

1. What is your gender?
    - Male
    - Female
    - Non-binary / third gender
    - Prefer not to say
    - Other

2. What is your age?

    [Open text field; numeric response between 0 and 150]

3. What is your race/ethnicity? (Select all that apply)

    - White / European
    - Black / African American
    - Hispanic / Latina
    - East Asian / South-East Asian
    - South Asian
    - West Asian / Middle-Eastern
    - American Indian / Indigenous / Alaska Native
    - Native Hawaiian / Other Pacific Islander
    - Other (Please Specify): _____

4. How do you identify politically?

    - Liberal
    - Centrist
    - Conservative

5. How would you describe your religious beliefs?

    - Agnostic
    - Atheist
    - Buddhist

- Catholic
- Christian (Protestant)
- Hindu
- Jewish
- Muslim
- Other (Please Specify): _____

6. In which country do you currently reside?

   [Drop-down list of countries]

Appendix D

General Introduction

Imagine that you are moving to a new home in a small village, where you will **live for the next 12 years**.

The village has a population of **1,000 people**.

The village is unusual because **the number and types of crimes are exactly the same every year.**

Each crime has a single attacker and a single victim.

Each year, the village experiences **some number** of **intentional murders OR intentional robberies.**

Every resident has an equal chance of being a victim of these crimes. Your chance of being a victim to these crimes does not change over time, and does not depend on who you are.

Here is a picture of the village:

For the remaining questions, please answer as if you are a person who is moving to a home in the village described earlier.

*Please rank the following types of crime based on how concerned you would be about needing to protect yourself.
(**1** = **Most concerned** about needing to protecting yourself, **2** = **Least concerned** about needing to protecting yourself)
There is no right or wrong answer.

- Murder ⠿
- Robbery ⠿

The village picture was generated by ChatGPT

Murder Rankings

As a resident of this village, it is important to understand the laws that tell how you are **legally allowed to protect yourself**.

Imagine that the village could have **one of two types of self-defense laws**:
- **Unrestricted self-defense law**: You are allowed to use **any level of force** you choose to defend yourself against an attacker. For example, you are allowed to kill an attacker with a firearm if they punched you.
- **Proportionate self-defense law**: You are allowed to defend yourself, but the **level of force you use must be proportionate to the threat** posed by the attacker. In other words, your response must match the level of force used against you. For example, if an attacker punches you, it would be illegal to stab them using a knife.

Under the **unrestricted self-defense law**, would it be legal to shoot your attacker with a firearm if they punched you?

- ◯ Yes
- ◯ No

*Please rank the following communities based on how much you would prefer to move there. (1 = Most preferred community, 6 = Least preferred community).

To rank these communities, **click and drag** the items.

In this case, we are focusing on **murder rate**.

Apart from the murder rate and the self-defense law, these communities are the same. For example, they all have 1,000 residents and each resident has an equal chance of being murdered.

- Community F: 4 INTENTIONAL MURDERS per year and PROPORTIONATE SELF-DEFENSE LAW.
- Community B: 3 INTENTIONAL MURDERS per year AND UNRESTRICTED SELF-DEFENSE LAW.
- Community C: 2 INTENTIONAL MURDERS per year and PROPORTIONATE SELF-DEFENSE LAW.
- Community E: 4 INTENTIONAL MURDERS per year and UNRESTRICTED SELF-DEFENSE LAW.
- Community D: 3 INTENTIONAL MURDERS per year and PROPORTIONATE SELF-DEFENSE LAW.
- Community A: 2 INTENTIONAL MURDERS per year AND UNRESTRICTED SELF-DEFENSE LAW.

Robbery Rankings

As a resident of this village, it is important to understand the laws that tell how you are **legally allowed to protect yourself**.

Imagine that the village could have **one of two types of self-defense laws**:
- **Unrestricted self-defense law**: You are allowed to use **any level of force** you choose to defend yourself against an attacker. For example, you are allowed to kill an attacker with a firearm if they punched you.
- **Proportionate self-defense law**: You are allowed to defend yourself, but the **level of force you use must be proportionate to the threat** posed by the attacker. In other words, your response must match the level of force used against you. For example, if an attacker punches you, it would be illegal to stab them using a knife.

*Under the **proportionate self-defense law**, would it be legal to stab your attacker with a knife if they punched you?

- ◯ No
- ◯ Yes

*Please rank the following communities based on how much you would prefer to move there. (1 = Most preferred community, 6 = Least preferred community).

To rank these communities, **click and drag** the items.

In this case, we are focusing on **robbery rate**.

Apart from the robbery rate and the self-defense law, these communities are the same. For example, they all have 1,000 residents and each resident has an equal chance of being robbed.

- Community F: 4 INTENTIONAL ROBBERIES per year and UNRESTRICTED SELF-DEFENSE LAW. ⠿
- Community A: 2 INTENTIONAL ROBBERIES per year and UNRESTRICTED SELF-DEFENSE LAW. ⠿
- Community C: 2 INTENTIONAL ROBBERIES per year and PROPORTIONATE SELF-DEFENSE LAW. ⠿
- Community B: 3 INTENTIONAL ROBBERIES per year and UNRESTRICTED SELF-DEFENSE LAW. ⠿
- Community E: 4 INTENTIONAL ROBBERIES per year and PROPORTIONATE SELF-DEFENSE LAW. ⠿
- Community D: 3 INTENTIONAL ROBBERIES per year and PROPORTIONATE SELF-DEFENSE LAW. ⠿

Appendix E

Mixed Logit Model

$$\mathbf{X}_i = \begin{bmatrix} \text{self-defence legislation} \\ \text{number of crimes} \end{bmatrix}, \qquad \mathbf{D}_n = \begin{bmatrix} \text{independent self-construal} \\ \text{interdependent self-construal} \\ \text{trust in the police, courts, and government} \\ \text{safety} \end{bmatrix}.$$

The utility of participant n for alternative i is specified as

$$U_{ni} = \boldsymbol{\beta}(\mathbf{D}_n)^\top \mathbf{X}_i + \epsilon_{ni},$$

where

$$\epsilon_{ni} \sim \text{i.i.d. standard Type I extreme value.}$$

We allow preference heterogeneity across participants by letting the coefficient vector depend on participant characteristics:

$$\boldsymbol{\beta}(\mathbf{D}_n) = \boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{D}_n + \boldsymbol{\Upsilon}_n\mathbf{D}_n + \boldsymbol{\nu}_n,$$

where

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}, \qquad \boldsymbol{\Gamma} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & \gamma_{14} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & \gamma_{24} \end{bmatrix},$$

and $\boldsymbol{\Upsilon}_n \in \mathbb{R}^{2\times 4}$ and $\boldsymbol{\nu}_n \in \mathbb{R}^{2\times 1}$ are participant n's random error matrices:

$$\boldsymbol{\Upsilon}_n = \begin{bmatrix} \upsilon_{n11} & \upsilon_{n12} & \upsilon_{n13} & \upsilon_{n14} \\ \upsilon_{n21} & \upsilon_{n22} & \upsilon_{n23} & \upsilon_{n24} \end{bmatrix}, \qquad \boldsymbol{\nu}_n = \begin{bmatrix} \nu_{n1} \\ \nu_{n2} \end{bmatrix},$$

Letting $\boldsymbol{\upsilon}_n = \mathrm{vec}(\boldsymbol{\Upsilon}_n^\top) \in \mathbb{R}^8$ denote the vectorisation of its entries, we assume

$$\boldsymbol{\upsilon}_n \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}), \qquad \boldsymbol{\Omega} = \mathrm{diag}(\omega_1^2, \ldots, \omega_8^2),$$

$$\boldsymbol{\nu}_n \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma} = \mathrm{diag}(\sigma_1^2, \sigma_2^2),$$

with $\boldsymbol{\upsilon}_n$, $\boldsymbol{\nu}_n$, and ϵ_{ni} mutually independent. The first two random coefficients are assumed to follow a normal distribution as they can, conceptually speaking, be positive or negative across participants.

$\boldsymbol{\Upsilon}_n$ captures the notion that demographic variables may shift preferences differently across participants, while $\boldsymbol{\nu}_n$ captures the idea that unobserved demographic characteristics may shift preferences for each participant n. Each element in $\boldsymbol{\Upsilon}_n$ and $\boldsymbol{\nu}_n$ has mean zero as they are the error equivalents of $\boldsymbol{\Gamma}$ and $\boldsymbol{\alpha}$, respectively. These random error terms are estimated by using simulation. In addition, notice that $\boldsymbol{\alpha} \in \mathbb{R}^2$, $\boldsymbol{\Gamma} \in \mathbb{R}^{2\times 4}$, and $\mathbf{D}_n \in \mathbb{R}^4$, so $\boldsymbol{\beta}(\mathbf{D}_n) \in \mathbb{R}^2$.

Substituting into the utility specification yields

$$\begin{aligned} U_{ni} &= \boldsymbol{\alpha}^\top \mathbf{X}_i + (\boldsymbol{\Gamma}\mathbf{D}_n)^\top \mathbf{X}_i + (\boldsymbol{\Upsilon}_n \mathbf{D}_n)^\top \mathbf{X}_i + \boldsymbol{\nu}_n^\top \mathbf{X}_i + \epsilon_{ni} \\ &= \boldsymbol{\alpha}^\top \mathbf{X}_i + \mathbf{D}_n^\top \boldsymbol{\Gamma}^\top \mathbf{X}_i + \mathbf{D}_n^\top \boldsymbol{\Upsilon}_n^\top \mathbf{X}_i + \boldsymbol{\nu}_n^\top \mathbf{X}_i + \epsilon_{ni}. \end{aligned}$$

This formulation allows both observed and unobserved participant characteristics to shift the utility of the community attributes.

Linear-in-parameters representation.

Define

$$\mathbf{Z}_{ni} = \begin{bmatrix} D_{n1}X_{i1} \\ D_{n1}X_{i2} \\ D_{n2}X_{i1} \\ D_{n2}X_{i2} \\ D_{n3}X_{i1} \\ D_{n3}X_{i2} \\ D_{n4}X_{i1} \\ D_{n4}X_{i2} \\ X_{i1} \\ X_{i2} \end{bmatrix} \in \mathbb{R}^{10}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \\ \theta_6 \\ \theta_7 \\ \theta_8 \\ \theta_9 \\ \theta_{10} \end{bmatrix} \in \mathbb{R}^{10}.$$

Then, utility can be written as

$$U_{ni} = \boldsymbol{\theta}^\top \mathbf{Z}_{ni} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{ni}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_i + \epsilon_{ni},$$

where $\mathbf{Z}_{ni}^* = (D_{n1}X_{i1},\, D_{n1}X_{i2},\, \ldots,\, D_{n4}X_{i1},\, D_{n4}X_{i2})^\top \in \mathbb{R}^8$ collects the interaction terms only, so that $\boldsymbol{\upsilon}_n^\top \mathbf{Z}_{ni}^* = \mathbf{D}_n^\top \boldsymbol{\Upsilon}_n^\top \mathbf{X}_i$.

The correspondence between $\boldsymbol{\theta}$ and the original parameters is

$$\theta_{2k-1} = \gamma_{1k}, \qquad \theta_{2k} = \gamma_{2k}, \qquad k = 1, \ldots, 4,$$

and

$$\theta_9 = \alpha_1, \qquad \theta_{10} = \alpha_2.$$

Ranking model

Suppose participant n is presented with a set of six communities

$$C_n = \{x_1, x_2, x_3, x_4, x_5, x_6\},$$

and provides a complete ranking

$$x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6.$$

The ranking is interpreted as a sequence of choices.

At stage m, the participant chooses item x^m from the remaining choice set

$$C_n^{x^m} = C_n \setminus \{x^1, \ldots, x^{m-1}\}.$$

Under the random utility model, the probability of choosing alternative i from a choice set C is

$$P(i \mid C, \boldsymbol{\theta}, \boldsymbol{\upsilon}_n, \boldsymbol{\nu}_n) = \frac{\exp\Big(\boldsymbol{\theta}^\top \mathbf{Z}_{ni} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{ni}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_i\Big)}{\sum_{j \in C} \exp\Big(\boldsymbol{\theta}^\top \mathbf{Z}_{nj} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{nj}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_j\Big)}.$$

Therefore, the probability of observing ranking $x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6$ is:

$$L_n\Big(x^1 \succ_n x^2 \succ_n x^3 \succ_n x^4 \succ_n x^5 \succ_n x^6 \mid \boldsymbol{\theta}, \boldsymbol{\upsilon}_n, \boldsymbol{\nu}_n\Big) = \prod_{m=1}^{5} \frac{\exp\Big(\boldsymbol{\theta}^\top \mathbf{Z}_{nx^m} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{nx^m}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_{x^m}\Big)}{\sum_{j \in C_n^{x^m}} \exp\Big(\boldsymbol{\theta}^\top \mathbf{Z}_{nj} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{nj}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_j\Big)}.$$

See Kilani, K. (2025) for the proof of this likelihood function.

We wanted to estimate the parameters $\boldsymbol{\omega} = \{\boldsymbol{\theta}, \boldsymbol{\Omega}, \boldsymbol{\Sigma}\}$. The choice probabilities are

$$P_{ni} = \int\int L_{ni}(\boldsymbol{\theta}, \boldsymbol{\upsilon}_n, \boldsymbol{\nu}_n)\, f(\boldsymbol{\upsilon}_n \mid \boldsymbol{\Omega})\, f(\boldsymbol{\nu}_n \mid \boldsymbol{\Sigma})\, d\boldsymbol{\upsilon}_n\, d\boldsymbol{\nu}_n,$$

where

$$L_{ni}(\boldsymbol{\theta}, \boldsymbol{\upsilon}_n, \boldsymbol{\nu}_n) = \frac{\exp\Big(\boldsymbol{\theta}^\top \mathbf{Z}_{ni} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{ni}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_i\Big)}{\sum_{j=1}^{J} \exp\Big(\boldsymbol{\theta}^\top \mathbf{Z}_{nj} + \boldsymbol{\upsilon}_n^\top \mathbf{Z}_{nj}^* + \boldsymbol{\nu}_n^\top \mathbf{X}_j\Big)}.$$

The probabilities are approximated through simulation for any given value of $\boldsymbol{\omega}$:

1. Draw values $\boldsymbol{\upsilon}_n^r \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$ and $\boldsymbol{\nu}_n^r \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, where $r = 1$ denotes the first draw.

2. Calculate the logit formula $L_{ni}(\boldsymbol{\theta}, \boldsymbol{\upsilon}_n^r, \boldsymbol{\nu}_n^r)$ with these draws.

3. Repeat steps 1 and 2 many times and average the results.

This average is the simulated probability:

$$\breve{P}_{ni} = \frac{1}{R} \sum_{r=1}^{R} L_{ni}(\boldsymbol{\upsilon}_n^r, \boldsymbol{\nu}_n^r).$$

The simulated probabilities are inserted into the log-likelihood function to give the simulated log-likelihood:

$$\mathcal{SLL} = \sum_{n=1}^{N} \sum_{j=1}^{J} d_{nj} \ln \breve{P}_{nj},$$

where $d_{nj} = 1$ if participant n chose alternative j and zero otherwise. The maximum simulated likelihood estimator (MSLE) is the value of $\boldsymbol{\omega}$ that maximizes $\mathcal{SLL}$:

$$\hat{\boldsymbol{\omega}} = \arg\max_{\boldsymbol{\omega}} \ \mathcal{SLL}.$$

, which is the $\boldsymbol{\omega}$ that we end up with.